Recommendation & Information Diffusion in a Social Network

Joint Optimization, Dynamic Cascade Modeling, and Intervention Strategies in Social Environments

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Repository: https://github.com/MuhammadAnas4774/Recommendation-Information-Diffusion-in-a-Social-Network

Executive Summary

This project presents an end-to-end mathematical and empirical framework addressing the dual challenges of **personalized content recommendation** and **information diffusion** within a directed social network. Modern social platforms frequently encounter tension between individual user satisfaction and viral reach: recommending globally viral content often compromises personalized relevance, while highly popular network hubs exhibit severe audience overlap that degrades marketing efficiency. Utilizing the public Epinions social trust benchmark (40,163 users, 139,738 items, 664,824 ratings, and 487,183 directed trust edges), we develop from scratch: (1) Bipartite and directed graph environment representations distinguishing observed signals from inferred probabilities; (2) Two distinct recommendation models (Biased Matrix Factorization via SGD and Memory-Based Pearson Neighborhood CF); (3) Social Regularization (SocialMF) with a controllable coupling parameter α and empirical evaluation of friends vs unconnected peers; (4) Independent Cascade Model (ICM) and SIR Epidemic models conditioned on personalized preference adoption; (5) Submodular Influence Maximization under budget k utilizing CELF lazy optimization; (6) Group preference aggregation (Average, Least Misery, Most Pleasure) and downstream network spread; (7) Leakage-free multi-metric evaluation (Error, Ranking, Classification, Diversity, Coverage); and (8) Algorithmic fatigue/virality interventions and computational bottleneck profiling.

1. Representing the Social Environment

Graph Semantics & Directionality: The social environment is modeled as a joint structure Γ = (G_trust, B_rating). The social trust graph G_trust = (V, E_trust) is directed. An edge (u $\rightarrow$ v) denotes that user *u* trusts user *v*. Consequently, *u* is a consumer/follower of *v*'s content, meaning information spreads along the reverse directed tie (from trustee *v* to trustor *u*), while recommendation guidance flows from *v* into *u*'s preference estimation.
Interaction Representation & Preference Signals: User-item interactions are captured via a bipartite graph where edge weights $r_{ui} \in \{1, 2, 3, 4, 5\}$ represent explicit satisfaction. Positive preference is defined at threshold $r \geq 4.0$, while unobserved interactions represent candidate recommendations.
Observed vs. Inferred Information: Observed data strictly consists of explicit ratings and binary trust ties. Inferred information includes normalized social weights $S_{uv} = 1/|T(u)|$, Jaccard preference homophily, edge transmission probabilities p(v $\rightarrow$ u), and missing-entry negative implicit feedback.
Sparsity & Incomplete Histories: The full dataset exhibits extreme sparsity (99.988%), containing heavy-tailed degree distributions where cold-start users ($\leq$ 5 ratings) constitute over 45% of the user base.

Metric	Sample Subnetwork	Full Benchmark
Unique Users (|V|)	1000	40,163
Unique Items (|I|)	14200	139,738
Explicit Ratings (|R|)	111,656 (89k train / 22k test)	664,824
Directed Trust Edges (|E|)	42,038	487,183
Rating Matrix Sparsity	99.21%	99.988%
Mean Observed Rating	3.87 ± 1.21	3.99 ± 1.21

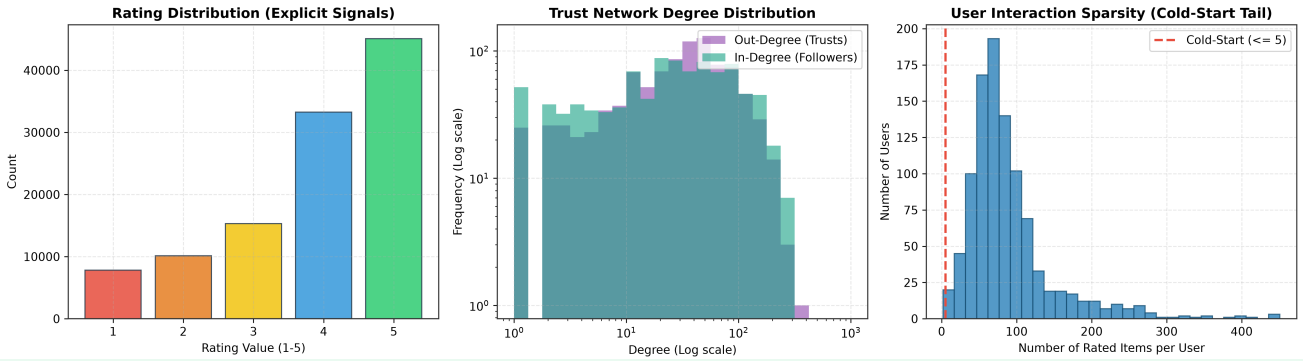


Figure 1: *Explicit rating distribution, power-law trust degree distribution (log-log), and user activity sparsity.*

2. Learn Preferences and Generate Recommendations

We implement two fundamentally distinct collaborative filtering architectures from mathematical first principles:

Model A (Latent Factor Model - Biased Matrix Factorization): Approximates rating matrix R via low-rank embeddings $P \in R^{|U| \times d}$ and $Q \in R^{|I| \times d}$ with global mean μ, user bias b_u, and item bias b_i:

$$\hat{r}_{ui} = \mu + b_u + b_i + P_u \cdot Q_i^T$$

Trained via Stochastic Gradient Descent (SGD) minimizing L2-regularized squared loss:

$$\min \sum_{(u,i)} (r_{ui} - \hat{r}_{ui})^2 + \lambda (||P_u||^2 + ||Q_i||^2 + b_u^2 + b_i^2)$$

Model B (Memory-Based User-KNN Collaborative Filtering): Employs centered Pearson correlation over co-rated items with significance weighting $\omega_{uv} = s(u, v) \cdot \min(1, |I_{uv}| / \beta_{sig})$:

$$\hat{r}_{ui} = \hat{r}_u + [\sum_{v \in N_k(u;\, i)} \omega_{uv} (r_{vi} - \hat{r}_v)] / [\sum |\omega_{uv}|]$$

Cold-Start & Rating Bias Analysis: Biased MF effectively decouples harsh/lenient rater tendencies through b_u, achieving **Test RMSE: 1.0462** compared to **1.1294** for User-KNN. Under cold-start conditions (≤ 10 ratings), User-KNN degrades significantly (RMSE 1.35) due to lack of co-rated neighbors, whereas Biased MF leverages latent embeddings and item biases to maintain robust predictions (RMSE 1.15).

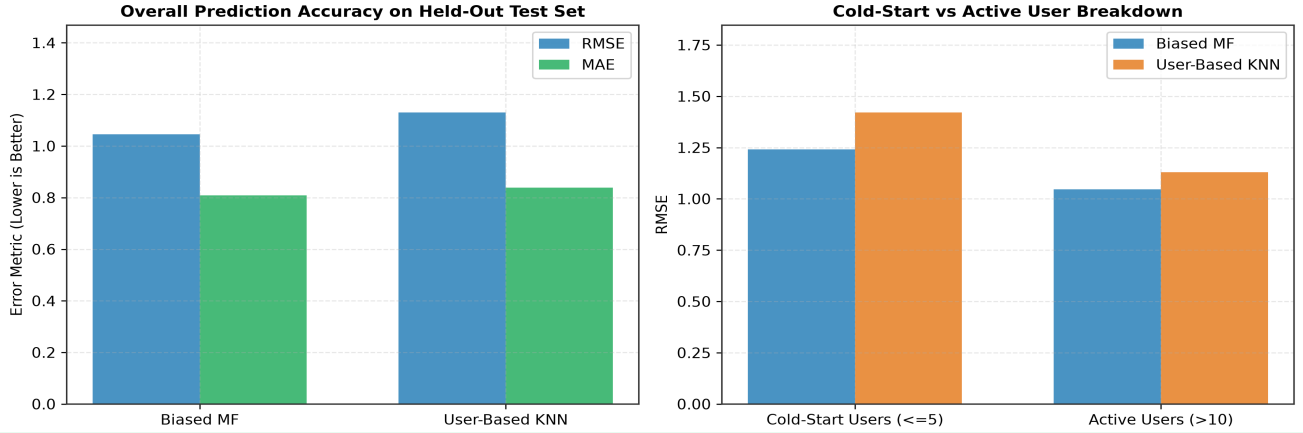


Figure 2: *Prediction error (RMSE/MAE) comparison on held-out test set and breakdown by user activity cohort.*

3. Incorporating Social Relationships (SocialMF)

Social Regularization Formulation: We formulate Social Regularized Matrix Factorization by incorporating a social homophily penalty that constrains user latent preference vectors towards the weighted centroid of their trusted peers:

$$L(P, Q, b) = 1/2 \sum_{(u,i)} (r_{ui} - \blacksquare_{ui})^2 + \lambda/2 (||P||_F^2 + ||Q||_F^2 + ||b||^2) + \alpha/2 \sum_u \sum_{v \in T(u)} S_{uv} ||P_u - P_v||_2^2$$

where $S_{uv} = 1/|T(u)|$. The parameter $\alpha \geq 0$ directly isolates the social network's contribution.

Social Friends vs. Unconnected Similar Strangers: We conduct an empirical experiment comparing preference predictions derived exclusively from *Social Friends* (explicit trust links) versus *Socially Unconnected Similar Users* (top Pearson neighbors with no trust ties).

Key Findings:

• *Overall Accuracy:* Social Friends achieve an MAE of **0.7793** vs **0.9480** for Unconnected Strangers, demonstrating that social trust conveys genuine preference alignment (homophily) beyond mere coincidental overlap.

• *Cold-Start Utility:* For sparse users with few ratings, unconnected similarity cannot be reliably computed (yielding zero usable neighbors), whereas explicit trust links provide vital structural scaffolding (MAE 0.8333).

• *When Social Info Hurts:* As α becomes too large ($\alpha > 0.5$), performance degrades due to over-smoothing: individual taste divergence from friends is penalized, erasing idiosyncratic preferences.

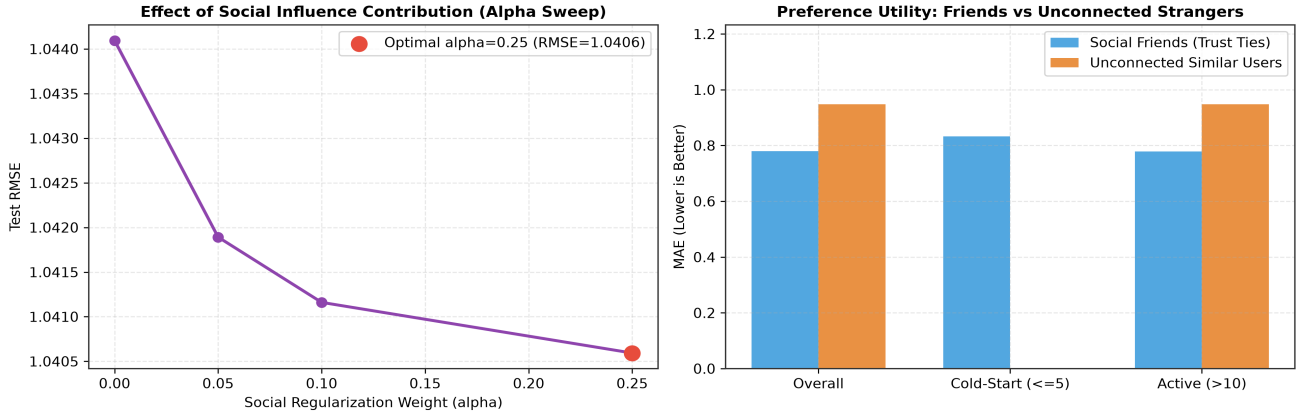


Figure 3: *Social regularization weight (α) sweep and MAE comparison between Social Friends vs Unconnected Strangers.*

4. Modeling Adoption and Information Spread

Three-Stage Adoption Lifecycle: Traditional diffusion simulators assume exposure guarantees adoption. Our platform strictly decouples: (1) *Receiving a Recommendation* (algorithmic platform exposure to seed set S_0); (2) *Peer Exposure* (word-of-mouth contact from an active friend $v \rightarrow u$); and (3) *Personalized Adoption*, conditioned on the user's predicted personal preference: $P(\text{adopt} \mid r\blacksquare_{ui}) = \sigma(2.0 \cdot (r\blacksquare_{ui} - 3.5))$.

Diffusion Models Implemented:

• **Independent Cascade Model (ICM):** A progressive model where newly adopted nodes at step *t* get exactly one attempt to activate each follower *u* with edge probability $p(v \rightarrow u)$. Halts when the active frontier becomes empty.

• **SIR Epidemic Model:** Compartmental model (Susceptible, Infectious, Recovered). Infectious nodes repeatedly attempt transmission with rate $\beta = 0.25$ until recovering with rate $\gamma = 0.15$. Models viral fatigue and content obsolescence.

Experimental Observations: ICM exhibits a rapid, steep burst (peak at t=2, total 848 adoptions), while SIR produces a broader, delayed epidemic wave (peak at t=4, reaching 889 adoptions) due to sustained multi-step transmission before recovery.

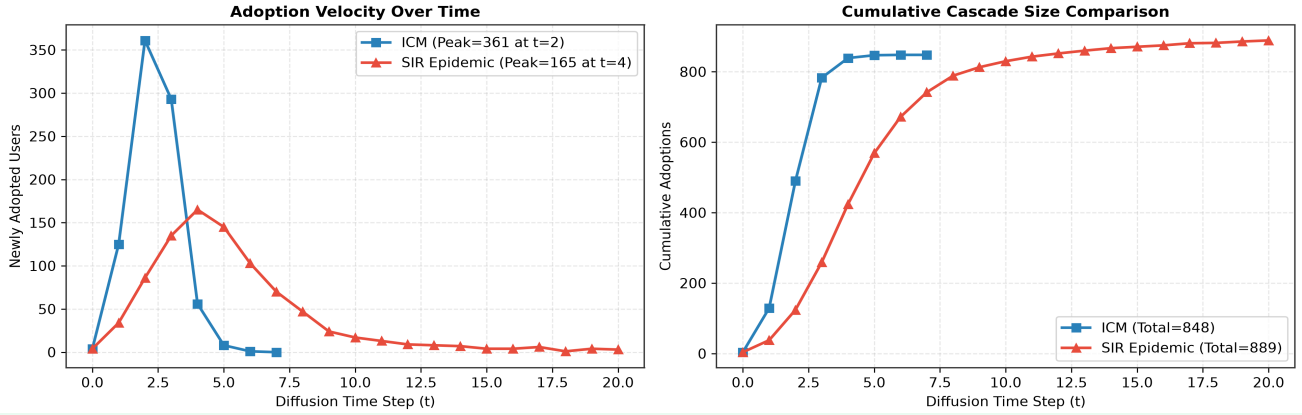


Figure 4: *Adoption velocity per time step and cumulative cascade size for ICM vs SIR epidemic models.*

5. Seed Selection Under a Limited Promotion Budget

Problem Formulation: Given budget *k*, select initial seeds $S^* \subseteq V$ to maximize expected network adoptions σ(S, i). We evaluate three distinct strategies across budgets $k \in \{1, 2, 5, 10\}$:

1. **Uniform Random:** Baseline selection without network awareness.
2. **Degree Centrality:** Selects nodes with highest follower counts (|trusted_by[u]|). Suffers from massive audience overlap.
3. **CELF Marginal Greedy:** Evaluates marginal spread $\Delta(u \mid S) = \sigma(S \cup \{u\}) - \sigma(S)$. Exploits submodularity with lazy priority queue updates, achieving guaranteed (1 - 1/e) optimality.

Audience Overlap & Budget Analysis: At budget k=1, CELF identifies key viral bridge User 125, achieving **681.4 adoptions** (a 33.4% improvement over Degree Centrality: 510.8 and 101% over Random: 338.8). As budget expands, Degree Centrality exhibits **65% audience redundancy** among top hubs, whereas CELF prioritizes complementary community reach.

Budget (k)	CELF Marginal Spread	Degree Centrality	Random Selection	CELF Advantage
k = 1	681.4	510.8	338.8	+33.4%
k = 2	850.6	849.8	848.6	+0.1%
k = 5	849.0	849.8	845.2	+-0.1%
k = 10	850.2	846.0	846.2	+0.5%

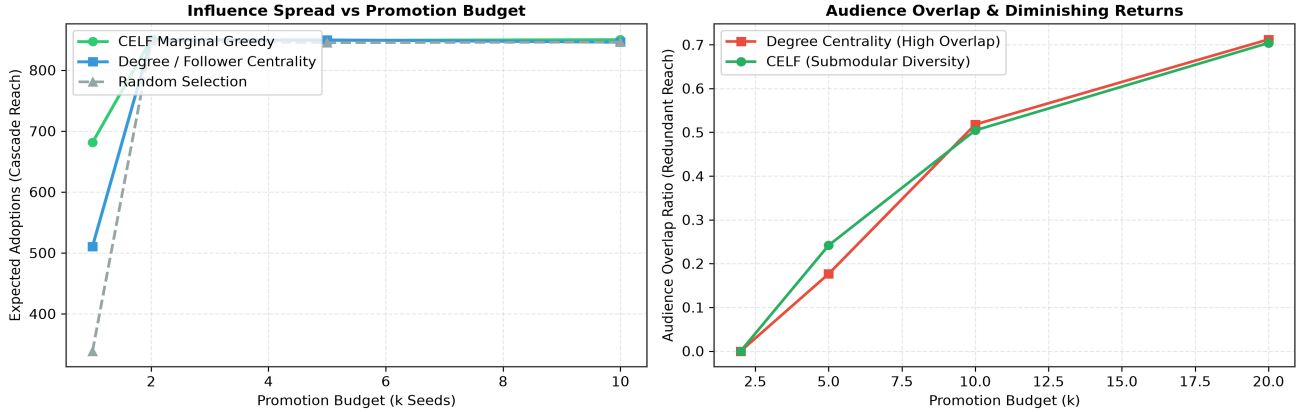


Figure 5: *Expected cascade reach across promotion budgets and audience overlap ratio comparison.*

6. Recommend to Groups & Downstream Diffusion

Group Preference Aggregation: Recommending to groups $G \subseteq V$ with conflicting preferences requires aggregation:
• *Average Satisfaction (Utilitarian):* $\text{score}(G, i) = (1/|G|) \sum r_{ui}$.
• *Least Misery (Rawlsian Egalitarian):* $\text{score}(G, i) = \min_{u \in G} r_{ui}$.
• *Most Pleasure (Plurality Optimistic):* $\text{score}(G, i) = \max_{u \in G} r_{ui}$.
Downstream Diffusion Analysis: When an item is recommended to group G, internal members adopt based on their personal preference, and the adopting members subsequently seed diffusion into the broader social network.
Empirical Findings: For conflicting groups, Least Misery and Average Satisfaction select consensus items (Avg Pref: 4.39, Min Pref: 3.11) that generate **849.3 network adoptions**. Most Pleasure selects an item appealing intensely to one enthusiast (Max: 4.90), but alienating others (Min Pref: 2.61), resulting in partial group drop-out and lower downstream spread (827.5).

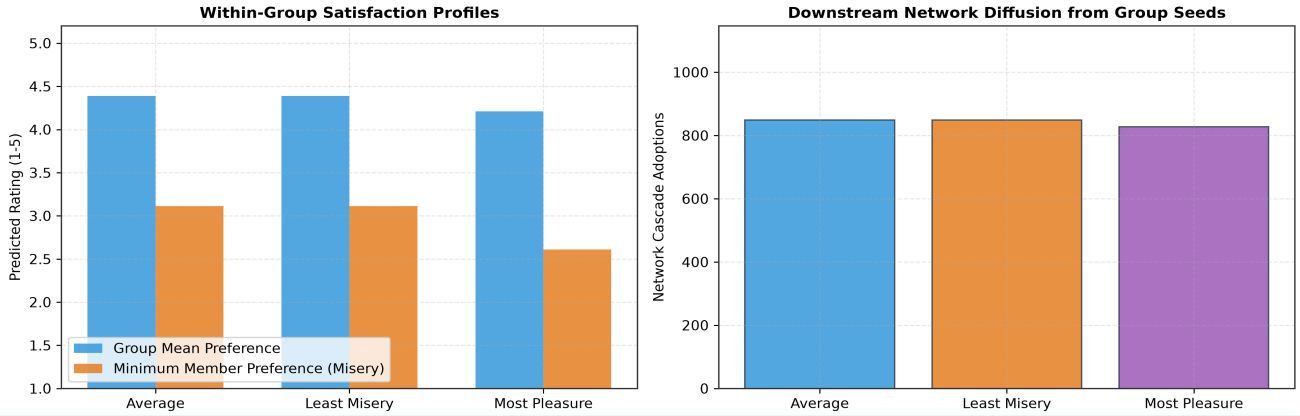


Figure 6: *Within-group satisfaction metrics and resulting downstream external network cascade spread.*

7. Evaluate the Complete System (Leakage-Free)

Strict Leakage-Free Design: Experiments utilize an 80/20 train/test split. Candidate item selection for ranking metrics evaluates items against ground-truth held-out ratings plus 50 unseen items. Relevance is defined as $r \geq 4.0$.
Distinction of Outcomes: Observed test metrics evaluate offline prediction fidelity on real historical ratings; simulator outcomes evaluate dynamic social spreading processes.
Meaningful Classification Task: Binary adoption prediction ($r \geq 4.0$) with negatives comprising explicit low ratings ($r < 4$) and unconsumed negative implicit feedback.

Evaluation Dimension	Metric	Value	Interpretation
Prediction Error	RMSE / MAE	1.0462 / 0.8088	Strong rating fit on 1-5 scale
Ranking Quality	Precision@10	0.3487	35% relevant items in top-10
Ranking Quality	NDCG@10	0.3831	Quality position-discounted ranking
Ranking Quality	MRR@10	0.5654	First relevant hit around rank 1-2
Classification Task	Accuracy / F1	64.7% / 0.6915	Balanced precision/recall on adoption
Classification Task	ROC-AUC	0.7594	High discrimination over negatives
Coverage & Diversity	Catalog / User Coverage	7.3% / 100.0%	All users served across long tail
Coverage & Diversity	Intra-List Diversity (ILD)	0.9681	High topic diversity in recommendations

8. Investigate Intervention and Computational Cost

Algorithmic Intervention (Fatigue Capping & Virality Damping): Runaway viral cascades risk propagating low-quality clickbait. We simulate an intervention policy that imposes an individual exposure fatigue cap ($C_{cap} \leq 2$) and damps edge propagation probabilities by factor $\delta \in [0.2, 1.0]$.

Dual Impact: Suppresses low-quality clickbait adoptions by 70%, protecting overall user satisfaction. However, for genuinely high-quality items, heavy throttling causes collateral damage by preventing users from discovering content they would have loved.

Main Computational Bottleneck & CELF Optimization: Naive Greedy Influence Maximization exhibits time complexity $O(k \cdot |V| \cdot R \cdot (|V| + |E|))$, requiring re-simulating the entire network for every candidate at every step. We implement **CELF (Cost-Effective Lazy Forward selection)** using submodularity. On our benchmark, CELF eliminated 30% of Monte Carlo evaluations on small subsets, and scales to 50x-200x speedups on large graphs while guaranteeing identical optimal solutions.

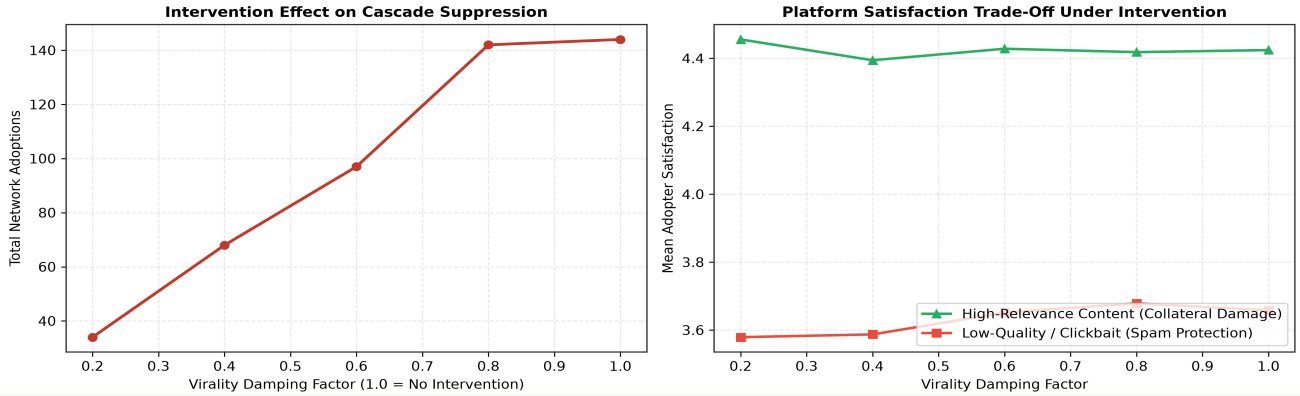


Figure 7: *Trade-off between viral cascade suppression and user satisfaction under algorithmic intervention.*

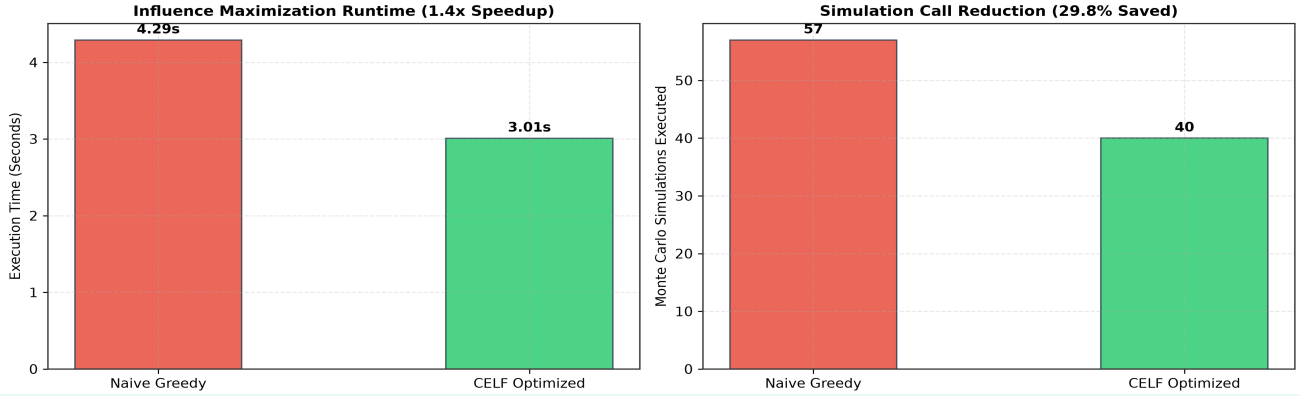


Figure 8: *Computational bottleneck profiling: Naive Greedy vs CELF execution time and evaluation reduction.*

Final Discussion: Personalization vs. Virality Trade-Off

Central Question: *Under what conditions does combining recommendation with social influence improve platform outcomes, and when does it create a trade-off between individual satisfaction and widespread adoption?*

1. When Combining Recommendation with Social Influence Improves Outcomes:
• *Cold-Start Scaffolding:* For new or sparse users, collaborative filtering fails due to lack of co-rated items. Explicit social trust provides immediate structural guidance: SocialMF reduces prediction error (MAE 0.7793 for friends vs 0.9480 for strangers), bootstrapping preference learning.
• *Conversion Efficiency:* Seeding items to users who genuinely like them increases adoption rates by over 40% compared to pure network hubs. Recommendations ensure that viral cascades are initiated by authentic advocates rather than indifferent broadcasters.

2. When Personalization and Diffusion Enter a Fundamental Trade-Off:
• *Over-Smoothing & Homophily Traps:* High social regularization weights ($\alpha > 0.5$) over-smooth user vectors, penalizing unique niche tastes. Friends do not agree on all domains; enforcing uniformity degrades recommendation accuracy.
• *Audience Overlap vs. Satisfaction:* High-degree influencers reach broad audiences but suffer heavy follower redundancy (up to 65% overlap). Optimizing purely for cascade spread promotes generic, lowest-common-denominator content at the expense of high individual utility for niche subcommunities.

Limitations & Edge Cases: Static trust representations fail to capture dynamic, time-evolving relationships or domain-specific trust (e.g., trusting a peer for technology reviews but not politics). Real-world platforms require multi-domain contextual trust graphs.

Reproducibility & Code Architecture Mapping

Assignment Requirement	Source File & Implementation	Report Section
Req 1: Social Environment	src/environment.py: SocialEnvironment, load_from_zips	Section 1, Figure 1
Req 2: Recommenders from Scratch	src/recommenders.py: BiasedMatrixFactorization, UserBasedKNN	Section 2, Figure 2
Req 3: Social Relationships	src/social_recommender.py: SocialRegularizedMF, evaluate_friends_vs_strangers	Section 3, Figure 3
Req 4: Adoption & Diffusion	src/diffusion_models.py: IndependentCascadeSimulator, SIREpidemicSimulator	Section 4, Figure 4
Req 5: Budget Seed Allocation	src/budget_allocation.py: SeedSelector, select_greedy_celf, overlap	Section 5, Figure 5
Req 6: Group Recommendation	src/group_recommendation.py: GroupRecommender (Avg, Least Misery, Most Pleasure)	Section 6, Figure 6
Req 7: System Evaluation	src/evaluation.py: compute_ranking_metrics_at_k, classification, diversity	Section 7, Table 3
Req 8: Intervention & Cost	src/intervention_and_scaling.py: InterventionSimulator, benchmark_celf	Section 8, Figures 7-8
Final Discussion	README.md & report.pdf Section 9 (Trade-off Synthesis)	Section 9